A Modular Unsupervised Framework for Attribute Recognition from Unstructured Text

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Abstract

We propose POSID, a modular and lightweight framework for extracting structured attribute–value descriptions from unstructured text without task-specific supervised training. The framework first identifies candidate text spans using lexical, semantic, or zero-shot candidate selectors and then applies a structured extraction procedure to recover attributes and their values. We use human attribute recognition in heterogeneous public-safety text as the primary case study and evaluate the approach on the InciText dataset. The present study focuses on extracting explicitly stated person attributes, including gender, race, height, and clothing descriptions, without training or fine-tuning on labeled InciText examples.

1 Introduction

Structured descriptions of people are frequently embedded in free-form text. An incident report, dispatch record, press release, missing-person notice, or other public-safety narrative may describe a person through attributes such as gender, race, height, build, clothing items, and clothing descriptors. These facts are useful only after they have been associated with the described person and converted from prose into a consistent structured representation. We study this problem as *attribute recognition from text*: given a document, first locate the text that contains relevant descriptions and then recover the corresponding attribute–value pairs.

The difficulty is not only that the input is unstructured. The source documents can differ substantially in genre, layout, sentence structure, abbreviations, and the location and density of attribute evidence. Some descriptions are compact (e.g., a demographic phrase followed by clothing); others distribute attributes across neighboring sentences or embed them in longer narratives. Fixed-vocabulary attributes such as gender can often be recognized with constrained lexical patterns, while variable-valued attributes such as clothing require recovering an open-ended set of item names and their modifiers. A single extraction strategy therefore has to cope with both variation in *where* useful evidence appears and variation in *how* an attribute is expressed.

A conventional approach could formulate the task as supervised sequence labeling, relation extraction, or question answering, provided enough labeled examples are available. Our setting instead asks how much structured information can be recovered *without task-specific supervised training*. Throughout this paper, we use *unsupervised* in this target-task sense: POSID is not trained or fine-tuned on labeled InciText examples. Individual pretrained components may themselves have been trained on external corpora or external tasks. In particular, one candidate selector uses a pretrained natural-language-inference model in a zero-shot configuration. This distinction is important: the framework avoids supervision for the attribute-recognition task while still permitting reusable pretrained representations.

We propose POSID as a modular two-stage framework. The first stage performs *candidate retrieval*: it identifies sentences or sentence fragments that are likely to contain relevant person descriptions. Candidate selectors may be lexical or semantic, including regular-expression matching, Word2Vec similarity [8], WordNet similarity [3], and zero-shot natural-language inference. The second stage performs *structured extraction*: it uses attribute-specific lexical constraints, local context, part-of-speech information, and lightweight semantic checks to recover attribute values and associate them with person mentions. Separating candidate retrieval from structured extraction makes the retrieval mechanism a replaceable front end rather than embedding one retrieval assumption throughout the extraction algorithm.

The separation is particularly useful for clothing descriptions. A precise lexical cue can strongly localize the relevant span, whereas a semantic candidate selector can recover descriptions that do not use the expected cue. After localization, however, not every noun or modifier in the candidate is necessarily part of a clothing description. This creates a second source of uncertainty inside the candidate itself: the extractor must decide when the local syntactic evidence remains sufficient to continue parsing and when it should stop. We therefore treat candidate relevance and extraction reliability as separate decisions rather than assuming that a semantically relevant sentence can be parsed exhaustively.

Human attribute recognition is used here as a concrete case study for this architecture. We evaluate POSID on InciText, which contains heterogeneous public-safety text and person-level annotations. The current 28-document historical subset is used as a small proof-of-concept evaluation spanning multiple source styles; the dataset and evaluation units are defined explicitly in Section 2. We do not use this small study to claim general cross-domain robustness. Instead, it establishes whether the proposed modular pipeline can recover structured attributes under the textual variation present in the available corpus.

The contributions of this paper are:

(1) We formulate person-attribute extraction from heterogeneous public-safety text as two separable stages—candidate localization and structured attribute recovery—and present POSID, a target-task-unsupervised framework in which lexical, distributional, and zero-shot semantic selectors feed a common POS-guided extractor for fixed attributes and open-vocabulary clothing descriptions.

(2) We conduct a controlled comparison of RE, Word2Vec, WordNet, zero-shot NLI, and routed combinations under the same

downstream recovery layer, allowing candidate-selection behavior to be studied without changing the structured extractor.

(3) We identify syntactic termination as a distinct extraction decision after candidate localization and empirically compare conservative early stopping with permissive continuation. The results expose a selector-dependent precision–recall tradeoff and show that locating a relevant candidate does not imply that all subsequent candidate text should be consumed as attribute evidence.

2 The InciText Dataset

InciText is a person-attribute text corpus assembled from public-safety documents and controlled text generation. The corpus is heterogeneous by construction: its sources differ in length, layout, writing style, degree of structure, and the location of person-description evidence. Because the experiments in this paper use only a small source-derived portion of the available data, we distinguish throughout this section between the *consolidated corpus inventory* and the *28-document proof-of-concept subset* used in the present evaluation.

2.1 Corpus Composition

The consolidated InciText inventory contains 558 documents across five subsets (Table 1). The 28-document source-derived portion comes from the West Lafayette Police Department (WLPD): ten short `Synthetic` descriptions provided by WLPD, thirteen WLPD press releases, and five WLPD incident/dispatch records. The synthetic descriptions were provided through WLPD and were written by investigating officers familiar with report writing to capture different ways in which person attributes may be expressed. The press releases are narrative-style public records, while the incident/dispatch material is more form-like and operational in structure. Together, these three subsets constitute the 28 documents used for the proof-of-concept experiments in this paper.

The consolidated inventory additionally contains 30 documents in the `Generated` subset and 500 in `Generated_flash`. These 530 records are free-form narratives created by the project's synthetic-data generator/augmentor using a fixed list of attribute values. They are distinct from the ten WLPD-provided `Synthetic` descriptions, which were written by domain professionals. The 530 generated records are retained in the corpus inventory but are not included in the 28-document evaluation reported here. A further 1,000 records from a `Generated_flash_2` source are not included in the consolidated 558-document corpus because standalone annotations are not available for that subset.

The distinction between these source types matters for attribute extraction. A compact synthetic description may place demographic and clothing attributes in one sentence, whereas a press release can distribute the same information across a longer narrative. Incident/dispatch records can introduce additional formatting, abbreviations, and sentence fragmentation. The dataset therefore provides a small but visibly mixed set of linguistic and document structures for testing the retrieval–extraction pipeline introduced in this paper.

Figure 1 summarizes these differences in source organization and the common person-level annotation target.

Table 1: Composition of the consolidated InciText corpus and the 28-document proof-of-concept evaluation subset. A dash indicates that the subset is not used in the proof-of-concept evaluation.

Source type	Consolidated	Proof-of-concept
Synthetic descriptions (WLPD)	10	10
Press releases	13	13
Incident / dispatch reports	5	5
Generated	30	–
Generated_flash	500	–
Total documents	558	28
Documents with person annotations	552	22
Person annotation records	553	23
Zero-person documents	6	6
One-person documents	551	21
Multi-person documents	1	1

2.2 Person-Level Annotation

Annotations are organized at the person level rather than only at the document level. Each document identifier maps to an array of person records, which allows a document to contain zero, one, or multiple annotated people without merging their attributes. In the 28-document subset, six documents contain no annotated person description, 21 contain one person record, and one document contains two person records, yielding 23 person-level annotation records in total.

Each person record contains a common structured schema covering demographic, physical, behavioral, and appearance attributes. The schema includes `gender`, `race`, `height`, `weight`, `hair_color`, `tattoos`, `backpack`, `headphones`, `smoking`, `posture`, `vehicle`, `facial_hair`, `hair_length`, and `build`. Clothing is represented by the nested `wearing` field. Rather than storing clothing as a flat label, `wearing` maps each item to zero or more descriptors. For example, a description containing a green jacket and blue pants is encoded as `jacket:[green]` and `pants:[blue]`. This representation lets the evaluation separately ask whether the clothing item itself was recovered and whether its associated descriptors were recovered.

The present paper evaluates the four attribute groups with sufficient coverage to support the proof-of-concept study: gender, race, height, and clothing. Table 2 reports their person-level gold coverage in the 28-document subset. Gender is present in all 23 person records, race in 22, clothing in 18, and height in 13. Other fields are considerably sparser in this subset: weight appears in seven person records and hair color in six, while each remaining populated field appears in at most three.

Missing attributes are retained explicitly in the annotation structure rather than interpreted as negative evidence. This is important for evaluation: absence of a gold value means that the source text does not provide an annotated value for that person, not that the corresponding physical or demographic property is known to be absent.

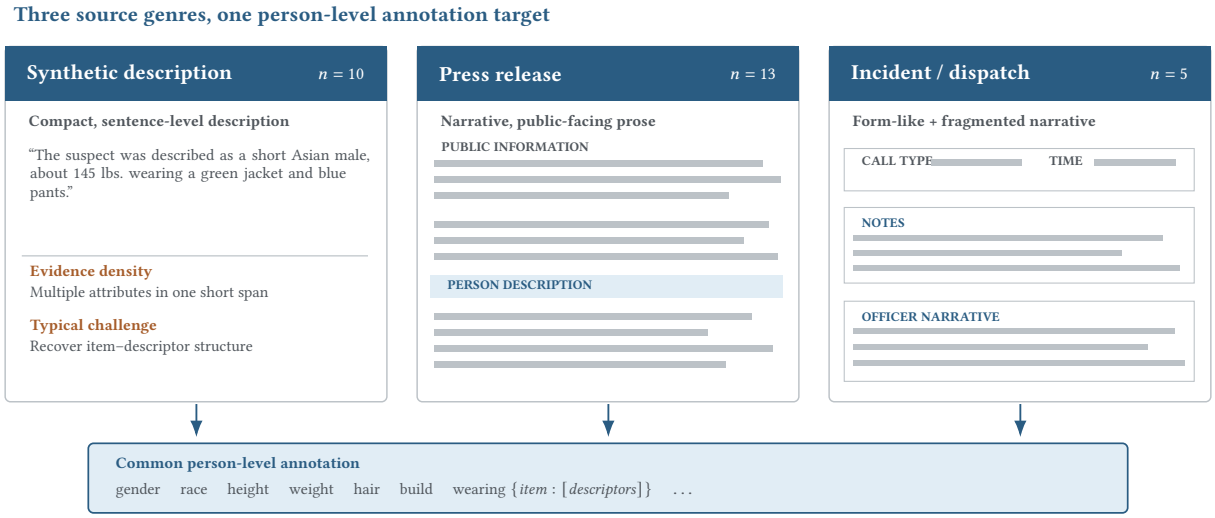


Figure 1: Source heterogeneity in the 28-document proof-of-concept subset. The synthetic panel shows an actual corpus example. The press-release and incident/dispatch panels schematically represent the organization of those source types rather than reproducing potentially identifying source text. Despite these differences in genre and layout, all three source types map to the same person-level annotation schema.

Table 2: Gold annotation coverage for the attribute groups evaluated in the 28-document proof-of-concept subset. Percentages use the 23 person-level annotation records as the denominator.

Attribute group	Person records	Coverage
Gender	23	100.0%
Race	22	95.7%
Height	13	56.5%
Clothing (`wearing`)	18	78.3%

2.3 Proof-of-Concept Evaluation Set

Unless stated otherwise, the experiments in this paper use the 28 source-derived documents consisting of the `Synthetic`, `Press Release`, and `Incident Reports` subsets. We use this set as a small proof of concept for structured attribute recovery across several document styles. The generated subsets are not mixed into the reported proof-of-concept results.

The evaluation preserves the person-level annotation structure. Consequently, a system can be evaluated not only on whether an attribute appears somewhere in a document, but also, where applicable, on whether the extracted attribute is associated with the correct person record. The exact matching rules for fixed attributes, clothing-item recovery, clothing descriptors, and person-aware scoring are defined in Section 4.

2.4 Provenance and Data Handling

The 28 source-derived records were obtained from or through WLPD. The ten synthetic descriptions were provided by WLPD; the incident/dispatch records are historical WLPD records; and the press releases are public-facing WLPD material. Identifying information in the historical incident-report material was redacted before research use. The consolidated metadata preserves the subset assignment, source path or lineage, and annotation linkage for each indexed document; event metadata is retained where available. Preserving this provenance is useful for reproducing the evaluation and for later analysis of source- or genre-specific behavior rather than flattening all documents into a single undifferentiated text collection.

The annotations represent attributes stated in the source text; POSID is evaluated on recovering those textual descriptions rather than predicting unobserved demographic or physical characteristics. The limitations of the annotation schema, source reporting practices, and demographic vocabulary are discussed separately in Section 6.

3 POSID Method

3.1 Problem Formulation and Overview

Let a document be an ordered sequence of sentences $T = \langle s_1, \ldots, s_n \rangle$. Given a small set of domain-relevant query phrases Q and a target attribute set O, the task is to recover a person-indexed collection of structured attribute–value pairs from T. In this work,

$$O = \{\textsc{gender}, \textsc{race}, \textsc{height}, \textsc{clothing}\},$$

where the first three attributes have constrained value forms and clothing is open-vocabulary: both the item name and its descriptors may vary with the text.

POSID decomposes this task into two stages (Figure 2). The first stage retrieves a small set of *candidate sentences* that are likely to contain relevant evidence. The second stage performs structured attribute extraction only over those candidates and then associates the resulting mentions with person records. This separation is intentional: a selector answers *where to look*, while the extraction

layer decides *what structured facts are sufficiently supported by the local text*. Different lexical or semantic selectors can therefore be compared without changing the downstream extraction procedure.

3.2 Candidate Representation

Every selector returns the same provenance-bearing representation

$$c = \langle i, x, r, q, \kappa \rangle,$$

where i is the sentence index, x is the text returned by the selector, r is its relevance score, q is the query or pattern responsible for the selection, and κ is the complete sentence or expanded local context. This representation is important because the selectors do not return exactly the same text. A regular-expression selector may return only the matched substring, whereas semantic selectors return the complete qualifying sentence. Keeping x and κ separately allows the extractor to preserve the selector's actual evidence while still recovering the context required for attribute association.

For sentence fragments beginning with a form of *wear* (e.g., "wearing a blue jacket"), the immediately preceding sentence is attached to κ when it contains an explicit person-role anchor such as *suspect*, *victim*, or *juvenile*. This handles a common fragmentation pattern in which the person is introduced in one sentence and the clothing description begins in the next.

3.3 Candidate Selectors

The candidate stage is implemented through a common selector interface. The four selectors used in this study differ in how they score a sentence and in how many candidates they return (Table 3). Let $\mathrm{tok}(s)$ denote the tokens of sentence s.

Pattern matching. For a query q, the RE selector uses exact lexical or regular-expression matching. Its relevance score is binary,

$$R_{\mathrm{RE}}(s, q) = \begin{cases} 1, & q \text{ matches } s, \\ 0, & \text{otherwise.} \end{cases}$$

Queries are checked in a fixed order and each sentence is emitted at most once; the first matching query becomes the candidate provenance q. This ordering is used later to distinguish candidates that can support clothing extraction from candidates intended only for fixed attributes.

Word2Vec similarity. For Word2Vec [8], candidate relevance is the strongest similarity between a sentence token and a query:

$$R_{\mathrm{W2V}}(s, q) = \max_{w \in \mathrm{tok}(s)} \frac{\mathbf{v}_w^\top \mathbf{v}_q}{\|\mathbf{v}_w\| \|\mathbf{v}_q\|}. \tag{1}$$

For multiple queries, the highest qualifying token–query pair supplies the candidate score and provenance. Tokens absent from the pretrained vocabulary are ignored.

WordNet similarity. WordNet [3] provides a lexical alternative to distributional embeddings. Noun senses are compared with Wu–Palmer similarity. In the configuration evaluated here, each query is represented by its first noun synset; sentence tokens are scanned in textual order and compared over their noun synsets. A sentence becomes a candidate when the first token–synset pair meeting the configured threshold is encountered. Thus the selector uses lexical-semantic proximity while retaining deterministic first-hit behavior.

Zero-shot NLI.. The fourth selector reformulates candidate retrieval as zero-shot text classification using a pretrained natural-language-inference model[12]. A sentence s is the premise and each query q is inserted into the hypothesis

$$h(q) = \text{"This example is } q\text{."} \tag{2}$$

The classifier's returned score for q is used as $R_{\mathrm{NLI}}(s, q)$. The highest-scoring query above the configured threshold supplies the candidate provenance. No InciText examples are used to train or fine-tune this selector; the pretrained NLI model is used in a zero-shot configuration. The exact checkpoint and thresholds are reported in Section 4.

3.4 Routed Candidate Selection

A central design choice is that combining selectors does not mean taking their unconditional union. The routed configuration first applies the high-precision RE selector using two query roles: a clothing cue based on *wearing* and a fixed-attribute pattern that localizes a suspect description. Let W denote RE candidates produced by the clothing cue and F the remaining RE candidates. A semantic selector M is invoked with the query *clothes* only when no clothing RE candidate is available:

$$C = F \cup \begin{cases} W, & W \neq \varnothing, \\ M(T, \{\text{clothes}\}), & W = \varnothing. \end{cases} \tag{3}$$

The semantic fallback can therefore recover an implicit clothing description without adding semantic candidates when the explicit cue already localized the relevant text.

Candidate provenance also controls downstream scope. A candidate produced by *wearing* is allowed to contribute both fixed attributes and clothing; a semantic *clothes* candidate contributes clothing only; and the fixed-attribute RE candidate contributes gender, race, and height only. Since the same sentence can legitimately enter through different roles, duplicate work is removed using the tuple (i, x, b_f, b_c), where b_f and b_c indicate whether fixed and clothing extraction are enabled. Two identical sentence strings are therefore not collapsed when they represent different extraction roles.

Algorithm 1 summarizes the routed pipeline. Standalone selector experiments use the same extraction stage but populate C directly from the corresponding selector rather than applying the fallback rule in Equation 3.

3.5 Structured Extraction of Fixed Attributes

Fixed attributes are extracted only from candidates whose provenance enables fixed-attribute processing. The extractor records an attribute mention with its sentence index, character position, and nearest local person anchor so that association can be resolved after all candidate evidence has been processed.

Gender. Gender is recognized through explicit whole-word lexical forms. The current mapping normalizes *male*, *man*, and *boy* to `male`, and *female*, *woman*, and *girl* to `female`. The system extracts only what is explicitly stated; it does not infer gender from a name.

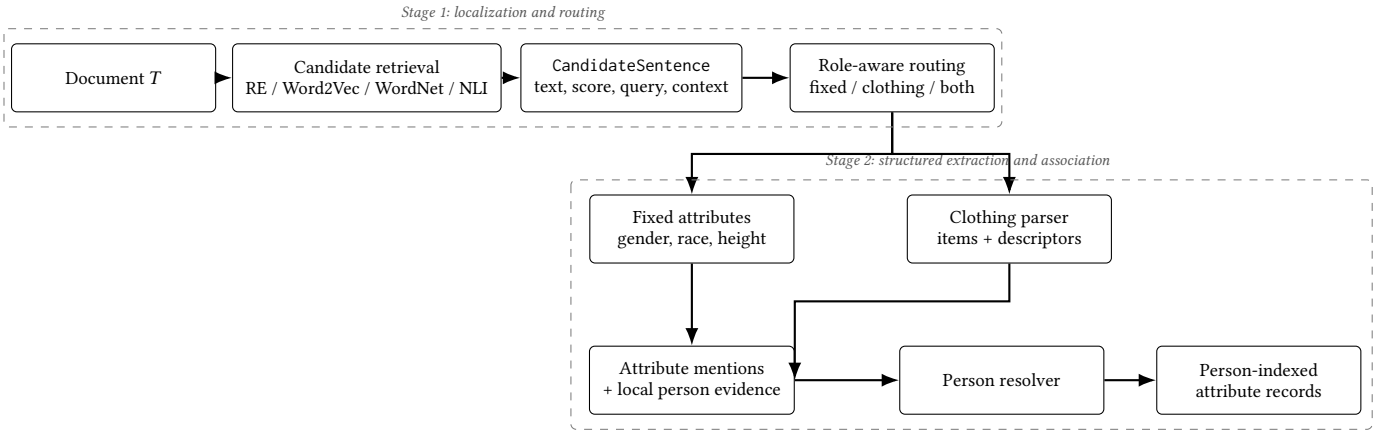


Figure 2: POSID separates candidate localization from structured extraction. Candidate provenance determines which extraction operations are valid for a candidate, while the final resolver associates extracted mentions with person records.

Table 3: Candidate-selector semantics used by POSID. Semantic selectors return the first sentence that satisfies their threshold in the current implementation; RE can return multiple matched sentences.

Selector	Relevance criterion	Returned text	Candidate cardinality
RE	Literal or explicit regular-expression match	Matched substring with full sentence in context	All matching sentences
Word2Vec	Maximum token–query cosine similarity	Full sentence	First qualifying sentence
WordNet	Noun-synset Wu–Palmer similarity	Full sentence	First qualifying sentence
Zero-shot NLI	Query-conditioned NLI label score	Full sentence	First qualifying sentence

Race. Race is extracted from explicit race–gender adjacency rather than from an isolated demographic token. For the race vocabulary

{asian, black, white, hispanic, latino, latina, indian},

the extractor accepts both orders, ⟨race, gender⟩ and ⟨gender, race⟩. Thus phrases such as "Asian male" and "male Asian" support the same race value. Requiring the local gender relation also prevents a color-like token occurring later in a clothing description from being interpreted as race.

Height. Numeric height expressions are recognized with patterns covering forms such as "6 feet 2 inches tall", "6-foot", "6 ft", and "5'10"". Qualitative *short*/*tall* is accepted only when it participates in a demographic construction containing gender, optionally with an intervening race term. This prevents expressions such as "short hair" or "tall building" from becoming height values. When a fixed-attribute candidate contains no height, POSID additionally inspects the immediately following sentence and retains only height mentions from that sentence. Person anchors are resolved normally, so height evidence attached to another explicitly named person is not reassigned to the current person.

3.6 Open-Vocabulary Clothing Extraction

Clothing is treated differently because the set of possible item names and modifiers is not closed. For each clothing-enabled candidate, POSID first defines the text span to parse. If a form of *wear* is present, the span begins at that cue and continues through the remainder of the sentence. If no wear cue is present—as can occur for a semantic fallback candidate—the extractor removes spans already identified as gender, race, or height and parses the remaining candidate text. This keeps the candidate selector responsible for localization while allowing implicit clothing constructions such as a list of apparel items without the word *wearing*.

The selected span is tokenized and part-of-speech tagged with NLTK [6]. The parser maintains a descriptor buffer D and a current noun run N. Its principal rules are:

(1) adjective tokens are accumulated as descriptors;
(2) noun tokens form open-vocabulary clothing item names, with consecutive nouns merged into compounds such as *tank top* or *dress pants*;
(3) a noun that is semantically close to the WordNet noun sense of *color* is treated as a descriptor rather than an item name, providing a safeguard for colors that are mistagged as nouns;
(4) contextual modifiers such as *dark* and *light* are retained as descriptors when they modify a following noun; and
(5) compound pre-nominal forms such as "buttoned up shirt" preserve the verb/particle modifier as a descriptor of the following clothing noun.

The resulting clothing fact is represented as $\langle item, D \rangle$. Surface forms are retained rather than mapped to a fixed clothing ontology, and

Algorithm 1: Candidate routing and structured POSID recovery

Input: document T; RE selector R; optional semantic selector M; parser mode τ
Output: person-indexed structured attributes $\widehat{A}$

$C_R \leftarrow R(T, \{q_{\text{wear}}, q_{\text{fixed}}\})$;
$W \leftarrow \{c \in C_R : c.q = q_{\text{wear}}\}$;
$F \leftarrow C_R \setminus W$;
if $W \neq \emptyset$ **then**
  $C \leftarrow F \cup W$;
else
  $C \leftarrow F \cup M(T, \{\text{clothes}\})$;
construct sentence-level person anchors, prior-anchor context, and shared-anchor context;
$L \leftarrow \emptyset$; $K \leftarrow \emptyset$; `// mentions and processed-role keys`
foreach $c \in C$ **do**
  $(b_f, b_c) \leftarrow \text{Role}(c.q)$;
  $u \leftarrow \text{Context}(c, T)$;
  $k \leftarrow (c.i, u, b_f, b_c)$;
  if $k \in K$ **then**
    continue;
  $K \leftarrow K \cup \{k\}$;
  if b_f **then**
    $L_f \leftarrow \text{Fixed}(u)$;
    $L \leftarrow L \cup L_f$;
    if L_f *contains no height and the next sentence exists* **then**
      $L \leftarrow L \cup \text{HeightOnly}(s_{c.i+1})$;
  if b_c **then**
    $L \leftarrow L \cup \text{ClothingParser}(u, \tau)$;
    ; `// Algorithm 2`
$\widehat{A} \leftarrow \text{ResolvePeople}(L)$;
return $\widehat{A}$;

Algorithm 2: Open-vocabulary clothing parser

Input: candidate context u; parser mode $\tau \in \{\text{CONTINUE}, \text{STOP}\}$
Output: clothing facts $S = \{\langle item, D\rangle\}$

$p \leftarrow \text{WearSpan}(u)$;
if $p \neq \emptyset$ **then**
  $x \leftarrow p$;
else
  $x \leftarrow \text{RemoveFixedSpans}(u)$;
$(w_1, t_1), \ldots, (w_m, t_m) \leftarrow \text{POSTag}(\text{Tokenize}(x))$;
$S \leftarrow \emptyset$; $D \leftarrow [\,]$; $N \leftarrow [\,]$;
Function `FLUSH()`**:**
  if $N \neq [\,]$ **then**
    $item \leftarrow \text{Join}(N)$;
    foreach $z \in \text{SplitSlash}(item)$ **do**
      $S \leftarrow S \cup \{\langle z, D\rangle\}$;
  $D \leftarrow [\,]$; $N \leftarrow [\,]$;
for $j \leftarrow 1$ **to** m **do**
  $w \leftarrow \text{casefold}(w_j)$; $t \leftarrow t_j$;
  if $t \in \{\text{VBD}, \text{VBN}\}$ *and* $t_{j+1} \in \{\text{RP}, \text{RB}\}$ *and* t_{j+2} *is noun* **then**
    append w w_{j+1} to D; continue;
  if $t \in \{\text{RP}, \text{RB}\}$ *and the previous token was consumed as a compound modifier* **then**
    continue;
  if w *is a boundary,* both, *or* but **then**
    `FLUSH()`; continue;
  if t *is adjective, or* $w \in \{$*dark, light*$\}$ *before a noun* **then**
    append w to D; continue;
  if t *is noun* **then**
    if w *is a person anchor or ignored scaffold word* **then**
      `FLUSH()`; continue;
    if $\text{ColorSim}(w) > \theta_{color}$ **then**
      `FLUSH()`; append w to D; continue;
    append w to N ; `// consecutive nouns form compound item names`
    continue;
  if t *is determiner, conjunction, preposition, punctuation, or delimiter* **then**
    `FLUSH()`; continue;
  if $\tau =$ *STOP and parsing has begun* **then**
    break;
  `FLUSH()` ; `// continue mode closes the current item and keeps scanning`
`FLUSH()`;
return S;

slash-separated item expressions are normalized into separate item records before output.

Unexpected POS patterns create a precision–recall decision. In the main parser, once clothing extraction has begun, an otherwise-unhandled POS pattern terminates the clothing scan. This conservative rule prevents candidate relevance from being treated as permission to consume arbitrarily distant nouns after the local syntactic evidence has become unreliable. A permissive continuation mode instead closes the current noun run and continues scanning, which can recover a later clothing mention at the risk of additional false positives. We retain continuation as an explicit ablation and evaluate the tradeoff in Section 5.

Algorithm 2 makes the open-vocabulary parser explicit. The procedure keeps the candidate-selection decision outside the parser: by the time it is called, the input has already been localized as clothing evidence. Within that evidence, the parser incrementally constructs item names and associated descriptors and decides whether syntactically unexpected material should merely close the current item or terminate the scan.

3.7 Person Association and Mention Resolution

Attribute extraction and person association are separated. Before candidates are processed, POSID scans the document for narrow person anchors: explicit roles (*suspect*, *victim*, and *juvenile*) and capitalized name-like tokens after excluding demographic and role stopwords. Anchors that occur inside the clothing portion of a sentence are not treated as person names, reducing collisions between apparel/color terms and person evidence.

Each extracted mention is first linked to the nearest preceding anchor in the same local context. The resolver then processes mentions in textual order and maintains the most recently active person records. Explicit anchors take precedence. When no explicit local anchor is available, pronoun cues (*he*, *she*, or *they*) carry recent person context into resolution. The resolver can use already extracted gender to prefer a compatible recent person for *he* or *she*; this is only an association constraint and never creates a new gender value. If a sentence introduces multiple anchors in a coordinated construction and a subsequent clothing sentence uses *both*, the clothing mention is copied to those shared anchors. Otherwise an unanchored mention is assigned to the nearest active local person, or to a deterministic anonymous person record if no person context exists.

The final output is a list of person records for each document. Fixed attributes are stored as scalar values, while clothing is stored as a list of item–descriptor structures. This person-indexed representation supports both document-level extraction metrics and the stricter person-aware diagnostic defined in Section 4.

4 Experimental Protocol

The experiments are designed to evaluate the two decisions separated by POSID: candidate localization and structured extraction. We first compare lexical and semantic candidate selectors while holding the downstream extractor fixed. We then report attribute-level extraction for the selected routed configuration. Finally, we isolate the clothing parser's behavior when local syntactic evidence becomes unreliable by comparing continued scanning against conservative early termination.

4.1 Evaluation Corpus

All experiments in this paper use the 28-document source-derived subset defined in Section 2.3: 10 synthetic descriptions, 13 press releases, and 5 incident/dispatch records. The subset contains 23 gold person-annotation records distributed across 22 documents; six documents contain no annotated person description and therefore also test whether a method introduces unsupported extractions. The gold annotations contain 23 gender values, 22 race values, and 13 height values. Clothing is represented as item–descriptor structures as described in Section 2.2.

Because POSID performs no target-task parameter learning, we do not create a supervised train/test split for this proof-of-concept study. Every method is run over the same 28 documents using the frozen selector queries, thresholds, and extraction rules listed below. The generated portions of InciText are not included in the results reported in this paper.

4.2 Compared Candidate-Selection Configurations

We evaluate four individual candidate selectors and two routed combinations. In every case, the selected candidate is passed to the same POSID structured extraction layer; the experiment therefore changes *where the extractor looks*, not the attribute extraction rules themselves. Table 4 summarizes the frozen configuration.

The RE selector is supplied two queries in a fixed order. The first is the literal cue `wearing`. The second is the fixed-attribute candidate pattern

```
[^.]*suspect.*male[^.]*\.
```

with case-insensitive matching. Query order is deliberate because a sentence matching both patterns is labeled by the first matching query and can therefore be routed as a clothing-bearing candidate while still permitting fixed attribute extraction. The routed methods retain fixed-attribute RE candidates regardless of whether the clothing path uses the RE candidate or the semantic fallback.

The Word2Vec selector uses the pruned `word2vec_sample` vectors available through NLTK. The WordNet selector represents `clothes` by its first noun synset and compares it against noun synsets of sentence tokens using Wu–Palmer similarity. The zero-shot selector uses `facebook/bart-large-mnli` through the Hugging Face `zero-shot-classification` pipeline with `multi_label=False`. For every sentence, the candidate label is inserted into the explicit hypothesis template

```
This example is {}.                (4)
```

including the terminal period. No target-corpus examples are used to train or fine-tune the NLI model.

4.3 Extraction Configuration

All candidate-selection conditions use the same POSID extractor described in Section 3. Sentence and word tokenization use NLTK, and clothing parsing uses the averaged-perceptron POS tagger. The WordNet color-sense safeguard treats a noun token as a color descriptor when its Wu–Palmer similarity to the first noun sense of *color* exceeds 0.70. Gender, race, numeric/relative height, next-sentence height recovery, compound clothing handling, slash-separated item handling, role-aware candidate processing, and person association are otherwise fixed across experimental conditions.

The main parser configuration sets `stop_on_unexpected_pos=True`. Once clothing parsing has begun, the first otherwise-unhandled POS token terminates the clothing scan. The controlled ablation changes only this flag to `False`: the current noun run is closed, but scanning continues and may recover a later item. The candidate selector, fixed-attribute rules, color threshold, tagging model, person-resolution logic, and all other parser behavior are held constant. The comparison therefore isolates the precision–recall effect of syntactic termination rather than replacing the parser with a different implementation.

Table 4: Candidate-selection configurations used in the proof-of-concept experiments. RE returns all matching sentences (at most one candidate per sentence); the semantic selectors return the first sentence satisfying the configured threshold.

Configuration	Query / evidence	Threshold	Selection or routing behavior
RE + POSID	`wearing`; fixed suspect pattern	exact match	All matching sentences; `wearing` is evaluated before the fixed pattern.
Word2Vec + POSID	`clothes`	≥ 0.50	First sentence whose best token–query cosine similarity qualifies.
WordNet + POSID	`clothes`	≥ 0.90	First qualifying sentence using Wu–Palmer noun-synset similarity.
Zero-shot NLI + POSID	`clothes`	> 0.85	First sentence whose zero-shot label score qualifies.
RE + WordNet + POSID	RE primary; `clothes` fallback	as above	Use RE clothing candidates when present; otherwise invoke WordNet for clothing.
RE + NLI + POSID	RE primary; `clothes` fallback	as above	Use RE clothing candidates when present; otherwise invoke zero-shot NLI for clothing.

4.4 Primary Evaluation Metrics

We report micro-aggregated precision, recall, and F_1 from counts accumulated across the evaluation documents:

$$P = \frac{TP}{TP + FP}, \qquad R = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2PR}{P + R}. \tag{5}$$

All textual values are compared after case folding and whitespace normalization. Height surfaces additionally undergo format normalization so that equivalent representations are compared consistently.

Fixed attributes. For gender, race, and height, gold and predicted values are pooled within each document and compared by exact one-to-one multiset matching after normalization. A matched value contributes one true positive; unmatched predictions and gold values contribute false positives and false negatives, respectively. Document-level counts are then summed before computing the corpus-level scores. This evaluation does not give credit for predicting an attribute that is not explicitly present in the gold annotation.

Clothing Attr-Only. Attr-Only evaluates recovery of clothing item names independently of their descriptors. Clothing instances are grouped by normalized item name and matched one-to-one within each document. A matched item contributes one true positive, while unmatched predicted and gold items contribute false positives and false negatives.

Clothing Attr-Value. Attr-Value evaluates the richer item–descriptor structure. Same-name clothing instances are paired one-to-one, choosing the pairing that maximizes descriptor overlap when duplicate item names occur. For each matched pair, the item name contributes one true-positive unit and every correctly recovered descriptor contributes an additional true-positive unit. Missing gold descriptors and extra predicted descriptors contribute false-negative and false-positive units. An entirely unmatched clothing item contributes one item unit plus its associated descriptor units to the corresponding error count. This metric therefore distinguishes partial descriptor recovery from an all-or-nothing item match.

4.5 Person-Aware Diagnostic

The primary metrics above measure whether the correct structured facts are recovered within each document. We additionally compute a stricter person-aware diagnostic to expose attribute-association errors. Gold and predicted people are paired deterministically, preferring an exact person ID, then an explicit matching role; singleton documents are paired directly. If multiple unresolved people remain, the pairing uses overlap in already extracted gender, race, height, and clothing-item evidence, with ambiguous ties recorded separately. Fixed attributes are then scored only within the matched person pair. For clothing, Attr-Only requires a matching item name within the person pair, while the strict person-aware Attr-Value diagnostic requires the matched item's descriptor set to agree exactly.

Because the present 28-document subset contains only one multi-person document, person-aware scores are treated as a diagnostic rather than evidence of broad multi-person generalization.

4.6 Reproducibility Configuration

The frozen runs use Python 3.9 with NLTK 3.9.2, gensim 4.4.0, NumPy 1.26.4, and SciPy 1.13.1. The zero-shot path uses PyTorch 2.2.2 and Transformers 4.44.2 with the `facebook/bart-large-mnli` checkpoint cached locally. The required NLTK resources are WordNet, `averaged_perceptron_tagger_eng`, and `word2vec_sample`. No model training or stochastic generation is performed during evaluation; every configuration operates on the same indexed document set and writes document-level predictions before metric computation.

5 Results and Analysis

We first report the frozen candidate-selection results with conservative early termination as the main clothing-parser configuration. We then examine precision and recall in more detail and use permissive continuation as a single-factor ablation. All results use the same 28-document proof-of-concept subset and the frozen settings in Section 4.

5.1 Candidate Selection and Attribute Recovery

Table 5 gives the overall F_1 view. The standalone Word2Vec, WordNet, and NLI selectors are configured only with the query `clothes`; fixed-attribute extraction is therefore outside their configured retrieval scope. We denote those cells by "–" rather than zero, because they are not attempted fixed-attribute conditions. RE-containing configurations expose the fixed-attribute candidate role in addition to the clothing path.

The fixed-attribute results are identical for RE, RE + WordNet, and RE + NLI because the semantic fallback is restricted to clothing retrieval. Rather than repeat the same precision/recall triplets for all three configurations, Table 6 reports them once. Precision is 1.0 for gender, race, and height, while recall is lower, indicating conservative fixed-attribute rules that avoid false positives on this subset but leave some annotated values unrecovered.

Clothing recovery is more sensitive to candidate selection. The detailed precision–recall values in Table 7 show that RE + WordNet obtains the highest main-configuration F_1 on both Attr-Only (.824) and Attr-Value (.857), with RE + NLI close behind (.814/.852). Standalone NLI is substantially more balanced than Word2Vec or WordNet, while standalone WordNet is precise when it fires but strongly recall-limited. These different retrieval profiles feed the same downstream POSID extractor, illustrating why candidate localization and structured extraction are treated as separate components.

5.2 Continuation Ablation

The main parser terminates once clothing extraction has begun and local POS evidence becomes unsupported. Table 8 asks what happens if this decision is reversed and scanning is allowed to continue. The ablation changes only this parser flag; fixed-attribute scores are therefore unchanged.

The largest effect occurs for RE + NLI. Attr-Only precision rises from .673 under continuation to .833 under the main stopping rule, while recall changes from .841 to .795; F_1 consequently increases from .747 to .814. Attr-Value shows the same pattern: precision increases from .796 to .904 while recall changes from .839 to .806, raising F_1 from .817 to .852. RE and RE + WordNet also improve under conservative termination, as does standalone NLI.

The direction reverses slightly for standalone Word2Vec and WordNet. This qualifies the result: stopping is not intrinsically superior for every retriever. Rather, it behaves as a precision-control rule when candidate localization has already placed POSID in a strongly relevant region. Permissive continuation can retain useful later evidence for weaker or sparse candidate selectors, but it can also admit unrelated nouns after the local syntactic signal has deteriorated. On this proof-of-concept set, the conservative policy provides the stronger tradeoff for the RE-based routed configurations used as the principal POSID setting.

5.3 Person-Aware Diagnostic

The evaluation protocol also defines a stricter person-aware diagnostic, but we do not mix its previously generated permissive-parser numbers into the main conservative-parser results. A person-aware rerun under the frozen main parser is required before reporting those values. This omission does not affect the primary document-level metrics above. Moreover, the present subset contains only one multi-person document, so any person-aware result from these 28 documents should in any case be interpreted as a diagnostic rather than as evidence of broad multi-person attribution capability.

6 Discussion and Limitations

6.1 Discussion

The experiments support a modular view of person-attribute extraction in which candidate retrieval and structured recovery solve related but distinct problems. Across the RE-containing configurations, gender, race, and height performance is unchanged when the clothing fallback selector changes. This is consistent with the intended routing architecture: the RE path supplies fixed-attribute evidence, while WordNet or zero-shot NLI is invoked only when an explicit clothing candidate is unavailable. The result is useful not because it establishes invariance across retrieval methods, but because it shows that changes to the clothing-retrieval front end need not perturb the fixed-attribute extraction path in the evaluated corpus.

A second finding concerns how much evidence the clothing parser should consume once a relevant region has been identified. POSID uses conservative termination: after clothing extraction has begun, an unexpected syntactic pattern terminates the scan rather than allowing subsequent tokens to be interpreted indefinitely as clothing evidence. The continuation ablation shows that this decision produces a genuine precision–recall tradeoff. Conservative termination improves clothing F_1 for RE, zero-shot NLI, and both routed RE+semantic configurations, with the largest gain observed for RE+NLI. In contrast, Word2Vec and WordNet alone lose slightly under early stopping. Despite these small standalone losses, conservative termination produces the highest clothing F_1 observed in the evaluation: .824 for Attr-Only and .857 for Attr-Value under RE + WordNet. This interaction suggests that parser behavior depends on candidate quality: when retrieval has already localized a strong descriptive region, continuing beyond locally supported syntax can introduce false positives; when retrieval is weaker or less precisely localized, continued scanning can occasionally recover useful later evidence. Early stopping should therefore be understood as an evidence-control mechanism rather than as an unconditional rule that is optimal for every retrieval strategy.

This observation also clarifies the role of the two stages of POSID. Candidate retrieval answers where relevant evidence may occur, but candidate relevance alone is not sufficient justification for treating every subsequent noun or modifier as part of the same attribute description. The structured extractor independently determines whether the local lexical and syntactic evidence remains compatible with the attribute being constructed. In this sense, the conservative parser prevents retrieval confidence from being implicitly converted into extraction confidence.

6.2 Limitations

Several limitations bound the interpretation of these results. First, the main evaluation contains only 28 WLPD-derived documents. They provide professionally grounded and heterogeneous public-safety evidence—ten WLPD-provided free-form synthetic descriptions written by domain professionals, thirteen press releases, and five incident/dispatch records—but they are insufficient to establish

Table 5: Primary F_1 results on the 28-document proof-of-concept subset with conservative early termination as the main clothing parser. "–" indicates that the standalone semantic selector was configured for clothing retrieval only, so fixed attributes are not applicable under that configuration.

Method	Gender	Race	Height	Clothing AO	Clothing AV
RE + POSID	.789	.811	.818	.795	.830
Word2Vec + POSID	–	–	–	.529	.540
WordNet + POSID	–	–	–	.414	.400
Zero-shot NLI + POSID	–	–	–	.741	.768
RE + WordNet + POSID	.789	.811	.818	**.824**	**.857**
RE + NLI + POSID	.789	.811	.818	.814	.852

Table 6: Fixed-attribute performance for all RE-containing configurations. The values are identical because semantic fallback changes only the clothing-retrieval path.

Attribute	Precision	Recall	F_1
Gender	1.000	.652	.789
Race	1.000	.682	.811
Height	1.000	.692	.818

broad robustness across agencies, domains, or report genres. We therefore treat the results as a proof of concept within the variation represented by this corpus.

The broader 558-document inventory should not be interpreted as equivalent external evidence. Its additional 530 `Generated` and `Generated_flash` records are free-form narratives produced programmatically by the project's synthetic-data generator/augmentor using a fixed inventory of attribute values, unlike the professionally authored WLPD `Synthetic` records. Full-corpus results are therefore supplementary to, rather than a substitute for, the WLPD-derived evaluation.

Person attribution is also only weakly tested because the present subset contains a single multi-person document. In addition, *unsupervised* denotes the absence of target-task supervised training on labeled InciText examples; pretrained components such as the NLI selector may themselves use external supervision. Finally, the framework does not eliminate bias: its outputs can inherit source-reporting, annotation-schema, lexicon, and missingness biases, including simplifications introduced by categorical demographic vocabularies.

7 Related Work

Earlier public-safety information-extraction systems established that suspect and crime attributes can be recovered from narrative text using domain rules, lexical resources, and NLP pipelines. Iriberri and Leroy developed a suspect description module that maps witness narratives into fields of a standard police report using lexical resources and hand-crafted semantic extraction rules [4]. Ku et al. broadened this setting to extract people, clothing, vehicles, weapons, locations, and time from police and witness narratives using a hybrid NLP pipeline [5]. More recent public-safety NLP work continues to emphasize the irregularity of police free text and the practical value of hybrid lexical, rule-based, and learned components when large domain-specific training sets are unavailable [1, 11].

POSID addresses a related extraction setting but separates the problem into candidate localization and structured attribute recovery rather than encoding the entire task in a monolithic set of domain rules. Candidate retrieval can be supplied by lexical, distributional, or zero-shot semantic mechanisms, while a common POS-guided extractor recovers fixed person attributes and open-vocabulary clothing descriptions. This separation also permits the retrieval and extraction stages to be evaluated independently and exposes an additional decision absent from earlier rule-based systems: when syntactic evidence is no longer sufficient to continue an extraction.

Recent zero-shot and schema-conditioned IE methods provide a different approach, using pretrained models to generalize extraction with little or no task-specific annotation. NLI has been used as a formulation for zero-shot text classification [12]; Sainz et al. cast relation extraction as entailment over verbalized relation labels [10]; and description- conditioned question answering has been used for zero-shot slot filling [2]. At a broader level, UIE learns a unified text-to-structure generator across multiple IE tasks and schemas [7], while GoLLIE uses annotation guidelines to improve zero-shot transfer to unseen IE tasks [9].

These approaches seek increasingly general learned extractors. POSID explores a different operating point: semantic models may be used to *locate* candidate evidence, but the final person-attribute representation is produced by an explicit structured recovery procedure rather than unconstrained generation. The goal is therefore not universal IE, but target-task-unsupervised and inspectable extraction from heterogeneous public-safety text.

The resulting positioning follows directly from the method and experiments. Earlier public-safety IE systems extracted suspect and crime attributes with domain rules and NLP pipelines, and POSID operates in the same problem family. Its architecture, however, makes candidate retrieval replaceable across RE, WordNet, Word2Vec, zero-shot NLI, and routed combinations while preserving a common POS-guided recovery layer. The final recovery step is therefore distinct from a generative IE model: zero-shot NLI is used as one candidate-selection mechanism, but the complete extraction task is not delegated to the pretrained model. The experiments further show that candidate selection and downstream extraction behavior are separable design decisions: conservative syntactic termination can improve final extraction even when the retrieved candidate is already relevant.

Table 7: Detailed clothing extraction results for the main conservative parser. AO denotes Attr-Only and AV denotes Attr-Value.

	Attr-Only			Attr-Value		
Method	P	R	F_1	P	R	F_1
Word2Vec + POSID	.750	.409	.529	.841	.398	.540
RE + POSID	.846	.750	.795	.910	.763	.830
WordNet + POSID	.857	.273	.414	.889	.258	.400
Zero-shot NLI + POSID	.811	.682	.741	.887	.677	.768
RE + WordNet + POSID	.854	.795	**.824**	.915	.806	**.857**
RE + NLI + POSID	.833	.795	.814	.904	.806	.852

Table 8: Main conservative parser versus permissive continuation. Δ is main F_1 minus continuation F_1; positive values favor conservative termination.

	Attr-Only F_1			Attr-Value F_1		
Method	Main stop	Continue	Δ	Main stop	Continue	Δ
Word2Vec + POSID	.529	.535	-.006	.540	.549	-.009
RE + POSID	.795	.761	+.034	.830	.813	+.017
WordNet + POSID	.414	.426	-.012	.400	.416	-.016
Zero-shot NLI + POSID	.741	.719	+.022	.768	.759	+.009
RE + WordNet + POSID	.824	.787	+.037	**.857**	.839	+.018
RE + NLI + POSID	.814	.747	**+.067**	.852	.817	**+.035**

8 Conclusion

We presented POSID, a target-task-unsupervised framework for structured person-attribute extraction from heterogeneous public-safety text. The framework separates candidate retrieval from structured recovery: lexical, distributional, and zero-shot semantic selectors can identify potentially relevant evidence, while a common POS-guided extraction layer recovers fixed person attributes and open-vocabulary clothing descriptions. This decomposition keeps the final extraction process explicit and inspectable without requiring labeled InciText examples for task-specific training.

The proof-of-concept evaluation shows that these stages remain distinct empirical decisions. RE-based routing provides stable recovery of the evaluated fixed attributes, while alternative semantic selectors can serve as replaceable clothing-retrieval mechanisms. Under the main conservative parser, RE + WordNet attains the highest observed clothing F_1, with .824 for Attr-Only and .857 for Attr-Value extraction. More importantly, the syntactic-termination experiment shows that retrieving a relevant candidate does not imply that the entire candidate should be consumed as attribute evidence: conservative termination improves final extraction for several retrieval configurations by limiting unsupported continuation, while producing a small loss for others. Candidate relevance and extraction reliability should therefore be treated as related but separate decisions.

These findings are intentionally limited to the variation represented in the current 28-document, professionally grounded WLPD-derived evaluation and do not establish broad cross-agency or cross-domain robustness. Larger independently sourced evaluations, stronger multi-person attribution tests, and analysis of the broader generated corpus remain necessary to characterize generalization. Within the present evidence, however, the central result is clear: lightweight candidate retrieval can remain replaceable while structured recovery retains explicit control over what evidence is extracted and when that evidence is no longer sufficient to continue.

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